Ex. No: 01
30.7.26

COMPUTER NETWORKS

1. BASIC NETWORKING COMMANDS

1. **arp -a**: Displays the ARP, cache, showing the IP addresses and corresponding MAC address of devices on the local network.

o/p - gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3

2. **hostname**: display the name of our computer

o/p: localhost.localdomain

3. **ipconfig/all**: Displays complete TCP/IP network configuration, including IP address, subnet mask, gateway, DNS, DHCP and MAC address.

o/p: Window IP configuration

Host Name . . . . . : ABIRAMI

Primary DNS suffix . . . . :

Node type . . . . . : Mixed

IP Routing Enabled . . . . : No

WINS proxy enabled . . . : No

4. **nbtstat -a**: Displays the NET BIOS name table of remote computer to help troubleshoot Net BIOS name resolution.

o/p: Displays protocol statistics and current TCP/IP connection using NBT.

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]

-a (adapter status) lists the remote machine's name table given its name.

-A (Adapter status) list the remote machine's name table given its IP address.

-n (names) lists local Net BIOS names.

5. netstat : Displays active network connections, listening ports and network statistics.

o/p: Active connections

Proto	Local Address	Foreign Address	State
TCP	[::1]:1521	ABIRAMI:49672	ESTABLISHED
TCP	[::1]:49672	ABIRAMI:1521	ESTABLISHED

6. netstat -r : Displays the routing table of the computer

o/p: Interface list

21...00 ff 2b da b2 66 .... TAP window adapter V9
6...3e 3d 3d 7f 5d 30 .... Realtek RTL8851BE Wifi 6 802.11ax PCIe Adapter #3
3...3a 8d 3d 7f 5d 30 .... Realtek RTL8851BE Wifi 6 802.11ax PCIe Adapter #5
2...38 3d 3d 7f 5d 30 .... Realttk RTL8851BE wifi 6 802.11ax PCIe Adapter
1..... Software loopback Interface 1.

7. nslookup : Finds the IP address of a domain name using the DNS server

o/p: Default server : unknown
Address : 127.0.0.1

8. pathping : Traces the route to a destination & measures packet loss at each network hop.

o/p: Usage : pathping [-g host-list] [-h maximum_hops]
[-i address] [-n] [-p period]
[-q num_queries] [-w timeout]
[-4] [-6] target_name

9. ping : Test connectivity b/w your computer & another computer or website using ICMP.

o/p: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL]
[-v TOS] [-r count] [-s count] [[-j host_list]
[-k host_list]] [-w timeout] [-R]
[-S srcaddr] [-c compartment] [-p]
[-4] [-6] target_name.

10. route: Displays the IP routing table used by the operating system to route network traffic.

o/p: ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METERIC metric]
[IF interface]

IMPORTANT LINUX NETWORKING COMMANDS:

1. ip <option><object><command>: The ip command is one of the basic commands. The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels

e.g: ip address show

o/p: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noque
state unknown group default qlen 1000
inet 127.0.0.1

2. ifconfig: The ifconfig command was/is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed

o/p: Widows IP configuration
unknown adapter McAfee VPN:
Media State . . . . . : Media disconnected
connection specific DNS suffix.

3. mtr: MTR is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute commands.

e.g: mtr google.com.

Host	Loss %	Snt	Last	Avg	Best
1. 172.16.8.1	0.0%	134	0.1	0.2	0.1
2. M2.97.163.129	2.6%	134	3.9	5.2	1.1
3. ???	2.2%	134	46.6	18.6	4.1

4. **tcpdump**: The tcpdump command is designed for capturing and displaying packets.

eg: dnf install -y tcpdump

tcpdump -D

O/P:

5. **ping**: Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet controll Message Protocol(ICMP) Echo request messages.

e.g: ping google.com.

O/P: Pinging google.com [142.250.67.46] with 32 bytes of data:
Reply from 142.250.67.46: bytes=32 time=9ms TTL=119
Reply from 142.250.67.46: bytes=32 time=3ms TTL=119
Reply from 142.250.67.46: bytes=32 time=3ms TTL=119
Reply from 142.250.67.46: bytes=32 time=3ms TTL=119

Ping statistics for 142.250.67.46:
Packets: Sent=4, Received=4, Lost=0(0% loss),
Approximate round trip times in milli-seconds:
Minimum=3ms, Maximum=9ms, Average=4ms.

Result:

Basic networking commands and linux commands are learned and executed.